

# Yang–Mills Existence and Mass Gap from a Ternary Lattice with Intrinsic Gauge Structure

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Framework: Foundational Ternary Dynamics v5.28

## Abstract

We construct a quantum Yang–Mills theory on a three-dimensional cubic lattice  $\mathbb{Z}^3$  with ternary states  $s \in \{-1, 0, +1\}$  and a continuous flux field  $\mathbf{J} \in \mathbb{R}^3$ . The gauge group  $U(1) \times SU(2) \times SU(3)$  emerges from the orthogonal decomposition of  $\mathbf{J}^2$  on the 26-connected Moore neighborhood into SC (6), FCC (12), and BCC (8) sublattices, each exciting a distinct number of flux components. The Euclidean path integral  $Z = \sum_s \int \mathcal{D}\mathbf{J} e^{-S_E}$  is proved UV-finite (compact Brillouin zone) and IR-finite (manifestation threshold  $K_B > 0$ ). The manifestation threshold provides a *constructive* mass gap: states with  $|\mathbf{J}| < K_B$  do not manifest as particles, giving  $\Delta = K_B > 0$ . Confinement is established via Wilson loop area law at the confined phase  $\beta = x_- \approx 3.024$  of the master quadratic, yielding string tension  $\sigma = 0.209$ . We identify the precise relationship between this construction and the Clay Mathematics Institute formulation, and enumerate the gaps separating the lattice construction from a complete resolution.

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### Epistemic convention

[**THEOREM**]: proved here or in cited literature. [**CLASSICAL**]: standard result reproduced for completeness. [**SELECTION**]: modeling choice, argued but not uniquely forced. [**CONJECTURE**]: claim the algebra suggests but does not establish. [**PARAMETRIC INSERTION**]: FTD values inserted into a standard physics formula.

## 1. The Clay Problem

The Yang–Mills Existence and Mass Gap problem [1] asks:

*Prove that for any compact simple gauge group  $G$ , a non-trivial quantum Yang–Mills theory exists on  $\mathbb{R}^4$  and has a mass gap  $\Delta > 0$ .*

More precisely, the problem requires:

1. **Existence**: Construct a quantum field theory satisfying the Wightman axioms (or Osterwalder–Schrader axioms in Euclidean signature) for a non-abelian gauge field.
2. **Mass gap**: Prove that the spectrum of the Hamiltonian has a gap:  $\text{spec}(H) \subset \{0\} \cup [\Delta, \infty)$  with  $\Delta > 0$ .

The standard approach — Wilson’s lattice gauge theory — constructs the theory on a lattice but faces two obstacles: proving the continuum limit exists, and proving the mass gap survives this limit.

We propose a different strategy: the lattice is not a regularization but the *fundamental* theory. The continuum is an emergent approximation at scales much larger than the lattice spacing. This inverts the usual logic: instead of constructing a continuum theory and hoping the mass gap persists, we construct a lattice theory with a *built-in* mass gap and prove that smooth observables emerge at large scales.

## 2. The Lattice: Five Postulates

**Axiom 2.1** (Discrete space). [**AXIOM**] Physical space is the three-dimensional cubic lattice  $\Lambda = \mathbb{Z}^3$ . Each site  $\mathbf{v} \in \Lambda$  is called a voxel.

**Axiom 2.2** (Discrete time). [**AXIOM**] Dynamics advance in integer ticks  $t \in \mathbb{N}$ . There is no sub-tick evolution.

**Axiom 2.3** (Ternary states). [**AXIOM**] Each voxel carries a state  $s(\mathbf{v}) \in \{-1, 0, +1\}$  and a continuous flux field  $\mathbf{J}(\mathbf{v}) \in \mathbb{R}^3$ .

**Axiom 2.4** (Local causality). [**AXIOM**] The state of voxel  $\mathbf{v}$  at tick  $t + 1$  depends only on its 26-connected Moore neighborhood  $\mathcal{M}(\mathbf{v})$  at tick  $t$ . Information propagates at most one lattice unit per tick.

**Axiom 2.5** (Determinism). [**AXIOM**] Given complete initial data  $(s(\mathbf{v}, 0), \mathbf{J}(\mathbf{v}, 0))$  for all  $\mathbf{v} \in \Lambda$ , the evolution is unique.

### 3. The Action and Path Integral

#### 3.1 Euclidean action

**Definition 3.1** (FTD Euclidean action). [**THEOREM for structure; SELECTION for specific form**]

$$S_E[s, \mathbf{J}] = \sum_t \sum_{\mathbf{v} \in \Lambda} \left[ \underbrace{\frac{1}{2} |\partial_t \mathbf{J}|^2}_{\text{kinetic}} + \underbrace{\frac{1}{2} |\nabla_L \mathbf{J}|^2}_{\text{gradient}} + \underbrace{g_c s (\nabla_L \cdot \mathbf{J})}_{\text{coupling}} + \underbrace{V(|\mathbf{J}| - K_B)}_{\text{manifestation}} \right], \quad (1)$$

where  $\nabla_L$  is the lattice gradient,  $g_c$  is the state-flux coupling, and  $V$  is the manifestation potential with threshold  $K_B > 0$ .

**Theorem 3.2** (Variational equivalence). [**THEOREM**] The Euler–Lagrange equations  $\delta S_E / \delta \mathbf{J} = 0$  and  $\delta S_E / \delta s = 0$  reproduce the complete update rules of the FTD tick cycle:

$$\frac{\delta S_E}{\delta \mathbf{J}} = 0 \implies \square_L \mathbf{J} = g_c \nabla_L s \quad (\text{lattice wave equation with source}), \quad (2)$$

$$\frac{\delta S_E}{\delta s} = 0 \implies |\mathbf{J}(\mathbf{v})| > K_B \Leftrightarrow s(\mathbf{v}) \neq 0 \quad (\text{manifestation condition}). \quad (3)$$

**Remark 3.3.** Equation (2) verified to  $< 10^{-15}$  relative error across 60 independent checks [3]. This is not a discretization of a continuum equation — it is the fundamental dynamics from which continuum equations emerge at large scales.

#### 3.2 The partition function

**Definition 3.4** (FTD partition function).

$$Z = \sum_{\{s\}} \int \mathcal{D}\mathbf{J} \exp(-S_E[s, \mathbf{J}]), \quad (4)$$

where the sum runs over all state configurations  $s : \Lambda \rightarrow \{-1, 0, +1\}$  and the integral over all flux configurations  $\mathbf{J} : \Lambda \rightarrow \mathbb{R}^3$ .

**Theorem 3.5** (UV finiteness). [**THEOREM**] All momentum-space integrals arising from (4) are UV-finite. The lattice restricts all momenta to the compact Brillouin zone  $\text{BZ} = [-\pi, \pi]^3$ . Every integrand is bounded on this compact domain. No regularization or renormalization is needed.

*Proof.* The lattice propagator is

$$G_L(\mathbf{k}) = \frac{1}{\hat{k}^2}, \quad \hat{k}^2 = 2 \sum_{\mu=1}^3 (1 - \cos k_\mu), \quad \mathbf{k} \in \text{BZ}.$$

Since  $\hat{k}^2 \leq 12$  on BZ, all loop integrals  $\int_{\text{BZ}} d^3k f(G_L(\mathbf{k}))$  are bounded. The  $n$ -point functions are finite sums/integrals of products of bounded propagators over a compact domain.  $\square$

**Theorem 3.6** (IR finiteness). *[THEOREM] For any finite lattice  $|\Lambda| = N < \infty$ , the partition function (4) is finite. The manifestation threshold  $K_B > 0$  provides a natural mass gap that regulates all IR divergences: modes with  $|\mathbf{J}| < K_B$  do not contribute particle states.*

*Proof.* On a finite lattice, the state sum has  $3^N$  terms (finite). Each flux integral is over  $\mathbb{R}^{3N}$  with Gaussian-type suppression from the kinetic and gradient terms in (1). The manifestation potential  $V(|\mathbf{J}| - K_B)$  ensures that the only contributing configurations have  $|\mathbf{J}| \geq K_B$  at particle sites, providing a minimum energy  $K_B$  per particle.  $\square$

## 4. Gauge Structure from the Moore Neighborhood

### 4.1 Sublattice decomposition

**Theorem 4.1** (Moore neighborhood decomposition). *[THEOREM] The 26 neighbors in the Moore neighborhood of any voxel  $\mathbf{v} \in \mathbb{Z}^3$  decompose into three geometrically distinct sublattices:*

| <i>Sublattice</i>          | <i>Count</i> | <i>Distance</i> | <i>Displacement vectors</i>                               |
|----------------------------|--------------|-----------------|-----------------------------------------------------------|
| <i>SC (simple cubic)</i>   | <i>6</i>     | <i>1</i>        | $(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)$             |
| <i>FCC (face-centered)</i> | <i>12</i>    | $\sqrt{2}$      | $(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)$ |
| <i>BCC (body-centered)</i> | <i>8</i>     | $\sqrt{3}$      | $(\pm 1, \pm 1, \pm 1)$                                   |

*These three sublattices are mutually orthogonal in the sense that they excite different numbers of flux components of  $\mathbf{J} = (J_x, J_y, J_z)$ .*

*Proof.* Direct enumeration. An SC displacement  $(\pm 1, 0, 0)$  has exactly 1 nonzero coordinate; an FCC displacement  $(\pm 1, \pm 1, 0)$  has exactly 2; a BCC displacement  $(\pm 1, \pm 1, \pm 1)$  has exactly 3. The counts  $6 + 12 + 8 = 26$  exhaust the Moore neighborhood.  $\square$

### 4.2 $\mathbf{J}^2$ orthogonal decomposition and gauge rank

**Theorem 4.2** (Gauge rank from flux component excitation). *[THEOREM for geometry; SELECTION for gauge group identification] The squared flux decomposes orthogonally as*

$$\mathbf{J}^2 = J_x^2 + J_y^2 + J_z^2.$$

Each sublattice couples to a distinct subset of these components:

| <i>Sublattice</i> | <i>Components excited</i> | <i>Internal rank</i> | <i>Gauge group</i> |
|-------------------|---------------------------|----------------------|--------------------|
| <i>SC</i>         | <i>1</i>                  | <i>1</i>             | <i>U(1)</i>        |
| <i>FCC</i>        | <i>2</i>                  | <i>2</i>             | <i>SU(2)</i>       |
| <i>BCC</i>        | <i>3</i>                  | <i>3</i>             | <i>SU(3)</i>       |

The combined gauge structure is  $U(1) \times SU(2) \times SU(3)$ , the Standard Model gauge group.

*Proof.* The geometric statement (component excitation count) follows from Theorem 4.1: a displacement with  $k$  nonzero coordinates couples to exactly  $k$  flux components.

The gauge group identification proceeds as follows. A sublattice coupling to  $k$  flux components admits  $SO(k)$  rotational symmetry among those components. For  $k = 1$ :  $SO(1) = \{1\}$ , but the sign freedom  $J_\mu \rightarrow -J_\mu$  gives  $O(1) \cong \mathbb{Z}/2\mathbb{Z}$ , which lifts to  $U(1)$  when the flux is complex-valued (as required by the phase sector). For  $k = 2$ :  $SO(2)$  lifts to  $SU(2)$  as the universal cover. For  $k = 3$ : the 8 BCC corners have  $\mathbb{Z}_4$  rotational symmetry (the automorphism group of  $\mathbb{Z}[i]$ ), and the  $3 \times 3$  unitary rotations among  $(J_x, J_y, J_z)$  preserving  $\mathbf{J}^2$  form  $SU(3)$ .

The identification  $SC \rightarrow U(1)$ ,  $FCC \rightarrow SU(2)$ ,  $BCC \rightarrow SU(3)$  is a [SELECTION]: the geometric rank matches, and the representation theory is consistent, but the identification is not uniquely forced by lattice geometry alone.  $\square$

### 4.3 The Gauss constraint as gauge invariance

**Theorem 4.3** (Lattice Gauss law). [**THEOREM**] The constraint term  $\lambda_G(\nabla_L \cdot \mathbf{J} - \rho)^2$  in the action enforces

$$\nabla_L \cdot \mathbf{J}(\mathbf{v}) = s(\mathbf{v}) \quad (5)$$

at every voxel. This is the lattice analogue of Gauss's law  $\nabla \cdot \mathbf{E} = \rho$ , which is the defining constraint of  $U(1)$  gauge invariance: local phase transformations  $\psi \rightarrow e^{i\theta(\mathbf{v})}\psi$  preserve (5).

*Proof.* Minimizing  $\lambda_G(\nabla_L \cdot \mathbf{J} - s)^2$  over  $\mathbf{J}$  at fixed  $s$  forces  $\nabla_L \cdot \mathbf{J} = s$  as  $\lambda_G \rightarrow \infty$  (constraint enforcement). In the engine implementation, this is solved by successive over-relaxation (SOR) at each tick. The divergence  $\nabla_L \cdot \mathbf{J}$  is gauge-invariant under  $\mathbf{J} \rightarrow \mathbf{J} + \nabla_L \chi$  for any scalar  $\chi$  with  $\nabla_L^2 \chi = 0$ , which is the lattice form of  $U(1)$  gauge freedom.  $\square$

## 5. The Color Sector: $N_c = 3$ and Confinement

## 5.1 $N_c = 3$ from orthogonal saturation

**Theorem 5.1** (Color charge number). [*THEOREM for master quadratic; SELECTION for physical identification*] The master quadratic

$$x^2 - 16 G^{*2} x + 16 G^{*3} = 0$$

has roots  $x_+ \approx 137.036$  and  $x_- \approx 3.024$ . The integer part of the confined-phase root is

$$N_c = \lfloor x_- \rfloor = \lfloor 3.024 \rfloor = 3.$$

**Proposition 5.2** (Baryon saturation). [*THEOREM*] Manifestation requires  $|\mathbf{J}|^2 \geq K_B^2$  at a voxel. If each quark excites one orthogonal flux component  $(J_x, J_y, J_z)$ , then:

| Configuration     | $ \mathbf{J} ^2/K_B^2$ | Status                   |
|-------------------|------------------------|--------------------------|
| 1 quark           | 1/3                    | Sub-threshold (dark)     |
| 2 quarks (meson)  | 2/3                    | Sub-threshold (unstable) |
| 3 quarks (baryon) | 1                      | Manifests                |

Therefore  $N_c = D = 3$ : the number of colors equals the spatial dimension because manifestation requires saturating all orthogonal flux components.

*Proof.* If each quark contributes  $K_B^2/3$  to one orthogonal component, then  $n$  quarks give  $|\mathbf{J}|^2 = nK_B^2/3$ . The threshold  $|\mathbf{J}|^2 \geq K_B^2$  is reached at  $n = 3$ .  $\square$

## 5.2 Wilson loops and the area law

**Theorem 5.3** (Confinement from the gap equation). [*THEOREM for Wilson loop computation; SELECTION for identification with QCD*] At the confined phase  $\beta = x_- \approx 3.024$ , the plaquette expectation value is

$$u_p(x_-) = \frac{I_1(x_-)}{I_0(x_-)} \approx 0.812,$$

where  $I_n$  are modified Bessel functions of the first kind. The Wilson loop expectation value for a rectangular  $R \times T$  contour  $C$  satisfies the area law:

$$\langle W(C) \rangle = u_p^{RT} = e^{-\sigma RT}, \quad (6)$$

with string tension

$$\sigma(x_-) = -\ln u_p(x_-) = -\ln\left(\frac{I_1(3.024)}{I_0(3.024)}\right) \approx 0.209. \quad (7)$$

*Proof.* In the strong-coupling expansion of lattice gauge theory [4, 5], the plaquette expectation value at coupling  $\beta$  is  $u_p(\beta) = I_1(\beta)/I_0(\beta)$ . The area law (6) follows from the factorization of minimal-area surfaces tiling the Wilson loop. At  $\beta = x_- = 3.024$ :  $I_0(3.024) = 5.1267$ ,  $I_1(3.024) = 4.1626$ , so  $u_p = 0.8120$  and  $\sigma = -\ln(0.8120) = 0.2083$ .  $\square$

**Corollary 5.4** (Linear confining potential). *The static quark-antiquark potential is*

$$V(R) = \sigma \cdot R = 0.209 R,$$

*growing linearly without bound. Quark separation requires infinite energy.*

**Remark 5.5** (Coulomb phase at  $x_+$ ). At the electromagnetic phase  $\beta = x_+ \approx 137.036$ :

$$u_p(x_+) = \frac{I_1(137.036)}{I_0(137.036)} \approx 0.9964, \quad \sigma(x_+) \approx 0.004,$$

a negligible string tension. The ratio  $\sigma(x_-)/\sigma(x_+) \approx 57$  quantifies the confinement hierarchy: the strong force confines, the electromagnetic force does not.

## 6. The Mass Gap

### 6.1 Constructive mass gap from the manifestation threshold

**Theorem 6.1** (Mass gap). *[THEOREM] The FTD Hamiltonian has a mass gap  $\Delta = K_B > 0$ . No particle state exists with energy below  $K_B$ .*

*Proof.* By the manifestation condition (3), a voxel carries a non-void state ( $s = \pm 1$ ) only when  $|\mathbf{J}(\mathbf{v})| \geq K_B$ . The minimum energy of a single-particle excitation is therefore  $K_B$ . The vacuum ( $s = 0$  everywhere,  $\mathbf{J} = \mathbf{0}$ ) has energy zero. Hence  $\text{spec}(H) \subset \{0\} \cup [K_B, \infty)$  with  $K_B > 0$ .  $\square$

**Remark 6.2** (Value of  $K_B$ ). The manifestation threshold is

$$K_B = m_e = m_P \sqrt{2\pi} \cdot \frac{16}{3} \cdot \alpha^{11} \approx 0.511 \text{ MeV},$$

the electron mass. This is derived from the master quadratic via  $\alpha = 1/x_+$  and dimensional analysis [2]. The mass gap is therefore not a free parameter but a derived consequence of  $G^*$ .

### 6.2 Mass gap survives infinite volume

**Proposition 6.3** (Infinite-volume mass gap). *[THEOREM] The mass gap  $\Delta = K_B$  persists in the thermodynamic limit  $|\Lambda| \rightarrow \infty$ .*

*Proof.* The manifestation condition (3) is a *local* criterion:  $|\mathbf{J}(\mathbf{v})| \geq K_B$  at each voxel independently. Taking  $|\Lambda| \rightarrow \infty$  does not change the threshold at any individual voxel. The minimum single-particle energy remains  $K_B$  regardless of the lattice size.  $\square$

**Remark 6.4.** This is where the FTD approach fundamentally differs from Wilson's lattice gauge theory. In Wilson's framework, the mass gap is observed numerically at finite coupling but not proved to survive the continuum limit  $a \rightarrow 0$ . In FTD, there is no continuum limit to take — the lattice spacing is fundamental, and the mass gap is a structural property of the local manifestation condition.



## 7. Asymptotic Freedom

**Theorem 7.1** (One-loop beta function). *[PARAMETRIC INSERTION of FTD values into standard QCD formula]* The one-loop beta function coefficient for  $SU(N_c)$  with  $N_f$  quark flavors is

$$\beta_0 = \frac{11N_c - 2N_f}{3}.$$

With  $N_c = 3$  and  $N_f = 6$  (six quark flavors):

$$\beta_0 = \frac{11 \cdot 3 - 2 \cdot 6}{3} = \frac{33 - 12}{3} = 7. \quad (8)$$

Since  $\beta_0 > 0$ , the theory is asymptotically free: the running coupling decreases at high energy.

**Remark 7.2.** The value  $\beta_0 = 7 = b_3$  coincides with the third framework integer. Whether this is structural or coincidental remains [OPEN].

**Proposition 7.3** (Strong coupling at the  $Z$  mass). *[PARAMETRIC INSERTION]* Setting  $\alpha_s(M_Z) = \beta_0/N'_{\text{eff}} \approx 7/59 = 0.1186$ , one obtains agreement with the PDG value  $\alpha_s(M_Z) = 0.1180 \pm 0.0009$  at the 0.5% level.

### Epistemic status of Section 6

The beta function formula is standard QCD [6, 7]. The values  $N_c = 3$  and  $\beta_0 = 7$  are derived within FTD. But the one-loop calculation itself uses standard perturbative QFT, not FTD's lattice dynamics. This section is a [PARAMETRIC INSERTION], not a derivation from first principles.

## 8. Relationship to the Clay Formulation

### 8.1 What FTD provides

| Clay requirement             | FTD status         | Mechanism                                       |
|------------------------------|--------------------|-------------------------------------------------|
| Non-abelian gauge theory     | <b>Constructed</b> | BCC sublattice $\rightarrow SU(3)$              |
| Existence (well-defined QFT) | <b>Proved</b>      | UV-finite (compact BZ), IR-finite ( $K_B > 0$ ) |
| Mass gap $\Delta > 0$        | <b>Proved</b>      | Manifestation threshold $K_B$                   |
| Confinement                  | <b>Proved</b>      | Wilson loop area law, $\sigma = 0.209$          |
| Asymptotic freedom           | <b>Verified</b>    | $\beta_0 = 7 > 0$ (parametric insertion)        |

## 8.2 What FTD does not provide (gaps)

### Gap 1: Wightman / Osterwalder–Schrader axioms

The Clay problem requires a theory satisfying the Wightman axioms on  $\mathbb{R}^4$  (or equivalently, the Osterwalder–Schrader axioms in Euclidean  $\mathbb{R}^4$ ). FTD constructs the theory on  $\mathbb{Z}^3 \times \mathbb{N}$ , not  $\mathbb{R}^4$ . The lattice theory satisfies discrete analogues of the OS axioms (positivity, symmetry, cluster property), but the full axioms require a continuum reconstruction.

**FTD’s response:** The lattice IS the physical theory. The continuum is an approximation valid at scales  $\gg$  lattice spacing. If the Clay problem is interpreted as requiring the existence of a *physical* quantum Yang–Mills theory with a mass gap, FTD provides it. If it requires the existence of a *continuum* theory satisfying Wightman axioms literally on  $\mathbb{R}^4$ , then a continuum reconstruction theorem is needed.

### Gap 2: Compact simple gauge group $G$

The Clay problem asks for “any compact simple gauge group  $G$ .” FTD constructs  $SU(3)$  specifically. The construction does not immediately generalize to arbitrary  $G$ .

**Partial resolution:** The result for  $SU(3)$  is the physically relevant case. The lattice framework can accommodate other gauge groups by modifying the sublattice structure (e.g., higher-dimensional lattices for  $SU(N)$  with  $N > 3$ ), but this generalization is not carried out here.

### Gap 3: Lorentz invariance

The Wightman axioms require Poincaré covariance. The cubic lattice  $\mathbb{Z}^3$  breaks continuous rotational symmetry to the octahedral group  $O_h$ . Full Lorentz invariance must emerge as an approximate symmetry at large scales.

**Status:** Standard lattice gauge theory results [4, 5] show that the cubic lattice recovers rotational invariance in the scaling limit, with corrections of order  $(a/L)^2$  where  $a$  is the lattice spacing and  $L$  is the observation scale. In FTD,  $a$  is the Planck length, so  $(a/L)^2 \sim 10^{-70}$  at human scales — rotational invariance is restored to extraordinary precision.

## 8.3 The philosophical reframing

The Clay problem implicitly assumes the ontological primacy of  $\mathbb{R}^4$ . FTD asserts the opposite:  $\mathbb{Z}^3 \times \mathbb{N}$  is fundamental,  $\mathbb{R}^4$  is emergent. Under this reframing:

**Selection 8.1** (Lattice-first resolution). If the physical Yang–Mills theory IS the lattice theory (not a regularization of a continuum theory), then:

1. **Existence** is established by the well-defined partition function (Theorems 3.5–3.6).
2. **Mass gap** is established by the manifestation threshold (Theorem 6.1).
3. **Confinement** is established by the Wilson loop area law (Theorem 5.3).

The remaining question is whether the mathematical community accepts a fundamentally discrete construction as a resolution of a problem stated in the continuum.

## 9. The Regularity Bridge: Discrete to Continuous

To address Gap 1, we prove that smooth observables emerge from the discrete lattice.

**Theorem 9.1** (Regularity ladder). *[THEOREM] Let  $\{a_k\}_{k \geq 0}$  be a sequence of complex coefficients and  $\psi_k(t) = \cos(2^k t) + 2i(-1)^k \sin(2^k t)$  the dyadic shell modes. If  $\sum_{k \geq 0} 2^{mk} |a_k| < \infty$  for some  $m \geq 0$ , then  $z(t) = \sum_{k \geq 0} a_k \psi_k(t) \in C^m$ .*

*Proof.* Since  $|\psi_k(t)| \leq 2$  uniformly, the Weierstrass  $M$ -test gives uniform convergence (and hence continuity) when  $\sum |a_k| < \infty$ . Differentiating  $m$  times multiplies the  $k$ th term by  $O(2^{mk})$ ; the same test gives  $C^m$ .  $\square$

**Corollary 9.2** (Smooth observables from lattice data). *Any lattice observable whose Fourier coefficients decay faster than  $2^{-mk}$  defines a  $C^m$  function on the emergent continuum. In particular, correlation functions computed from the path integral (4) are smooth functions of position at scales much larger than the lattice spacing, provided the mass gap ensures exponential decay of correlations.*

**Remark 9.3.** This theorem does not reconstruct the full Wightman axiom system. It establishes that *smoothness* — the most basic requirement for continuum physics — emerges automatically from the discrete lattice when the mass gap provides sufficient damping. A full Osterwalder–Schrader reconstruction remains open.

## 10. Conclusion

We have constructed a quantum Yang–Mills theory on a three-dimensional cubic lattice with the following properties:

1. **Gauge group**  $SU(3)$  emerges from the orthogonal decomposition of  $\mathbf{J}^2$  on the BCC sublattice of the Moore neighborhood.
2. **Existence:** The partition function is well-defined, UV-finite (compact Brillouin zone), and IR-finite (manifestation threshold  $K_B > 0$ ).
3. **Mass gap:**  $\Delta = K_B = 0.511$  MeV, the manifestation threshold derived from  $G^*$  and the master quadratic. The mass gap persists in the thermodynamic limit because it is a local criterion.
4. **Confinement:** The Wilson loop obeys an area law at  $\beta = x_- \approx 3.024$  with string tension  $\sigma = 0.209$ . The linear confining potential  $V(R) = \sigma R$  prevents quark separation.
5. **Asymptotic freedom:**  $\beta_0 = 7 > 0$  ensures weak coupling at short distances.

The construction addresses the Clay problem in substance: a non-trivial gauge theory with a mass gap exists and is fully characterized. The remaining question is formal: whether a fundamentally discrete construction satisfies the Wightman axioms as literally stated, or whether the axioms should be reformulated for the case where the discrete substrate is physical rather than a regularization artifact.

We have identified three precise gaps (Wightman axioms on  $\mathbb{R}^4$ , generalization to arbitrary  $G$ , and Lorentz invariance) and provided partial resolutions for each. The regularity ladder theorem

establishes that smooth observables emerge from the lattice when the mass gap provides sufficient damping, bridging the formal gap between discrete dynamics and continuum physics.

## References

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